

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and optimized distributed **AI model pre-training pipelines** across large-scale GPU infrastructures, reducing training time by **15%** through advanced parallel training strategies.
- Developed and iterated on novel **transformer architectures** for large language models, improving performance metrics by **10%** via rigorous experimentation and analysis.
- Identified and resolved computational bottlenecks in **large-scale AI training systems**, enhancing model scalability and achieving a **20%** reduction in average processing latency.
- Contributed to data curation strategies and baseline improvements for **multi-modal AI systems**, strengthening model training outcomes and increasing data utilization efficiency by **25%**.
- Enhanced **distributed training systems** to ensure seamless scalability and operational efficiency, supporting environments with over **1,000 GPUs**.
- Collaborated with cross-functional engineering and research teams to accelerate innovation in AI capabilities, contributing to the successful launch of **2 major pre-training initiatives**.

SKILLS

AI & Machine Learning – Large Language Models, Transformer Architectures, AI Pre-training, PyTorch, TensorFlow, JAX, Deep Learning Optimization, Model Scaling, Performance Tuning, Distributed Machine Learning Systems, GPU Training

Programming & Development – Python, Java, Spring Boot, REST APIs, Docker, Linux, Shell Scripting, Git, Jenkins

Data Engineering & Analytics – SQL, PostgreSQL, MySQL, Oracle, Data Warehousing, ETL, Pandas, NumPy, R, Power BI, DAX, Power Query

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Distributed Computing, Infrastructure Optimization

Research & Experimentation – Experimental Design, Data Curation, Bottleneck Analysis, Scalability Solutions, Research Mindset, Novel Techniques

Collaboration & Communication – Cross-functional Teams, Remote Collaboration, International Teams, Agile/Scrum Methodologies

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India

01/2021

PROJECTS

Distributed LLM Pre-training Framework

- Architected and implemented a scalable distributed framework for **large language model pre-training**, enabling the training of models with over 100 billion parameters on a cluster of **500 GPUs**.
- Integrated **PyTorch DistributedDataParallel (DDP)** and **NVIDIA NCCL** for efficient inter-GPU communication, achieving **90%** GPU utilization and reducing training epochs by **30%**.
- Developed automated data loading pipelines using **Python and Pandas**, processing terabytes of text data and optimizing data throughput to minimize idle GPU time, resulting in a **20%** faster data ingestion rate.

HCL Payments Operations Dashboard Enhancement

- Enhanced the **payments operations dashboard** by developing new Java Spring Boot microservices to monitor transaction flows in real-time, reducing critical alert response times by **25%**.
- Automated reconciliation workflows using **Python scripting** and SQL queries against PostgreSQL, improving data accuracy and reducing manual reconciliation effort by **40%**.
- Implemented comprehensive logging and monitoring using **Splunk**, which decreased the mean time to identify production issues (MTTI) by **35%** for financial-services transactions.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)